GNR644 REMOTE SENSING FOR SURFACE WATER RESOURCES

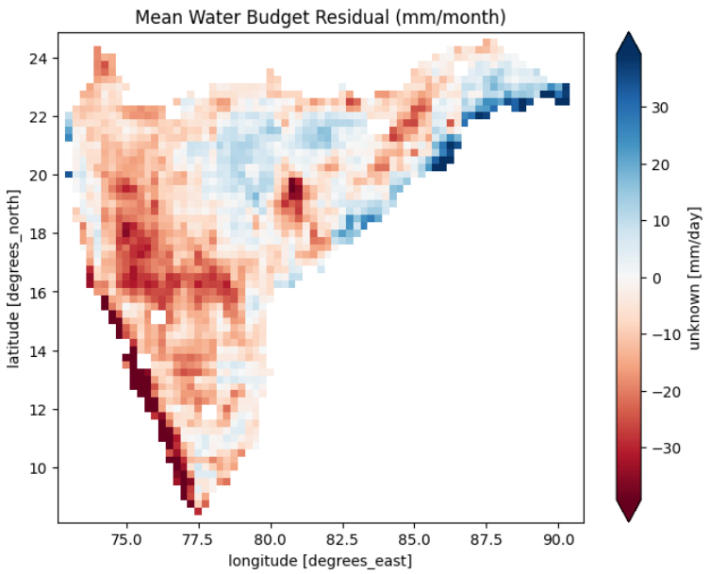
Group 2	
ROLL NUMBER	**NAME**
25M0304	SATYENDRANATH KAR
25M0611	RISHI GANESH L
25M0612	JAGU MANISH
25M0614	MINTI JASAKIYA

STUDY AREA: Peninsular India

WATER BUDGET RESIDUAL ANALYSIS

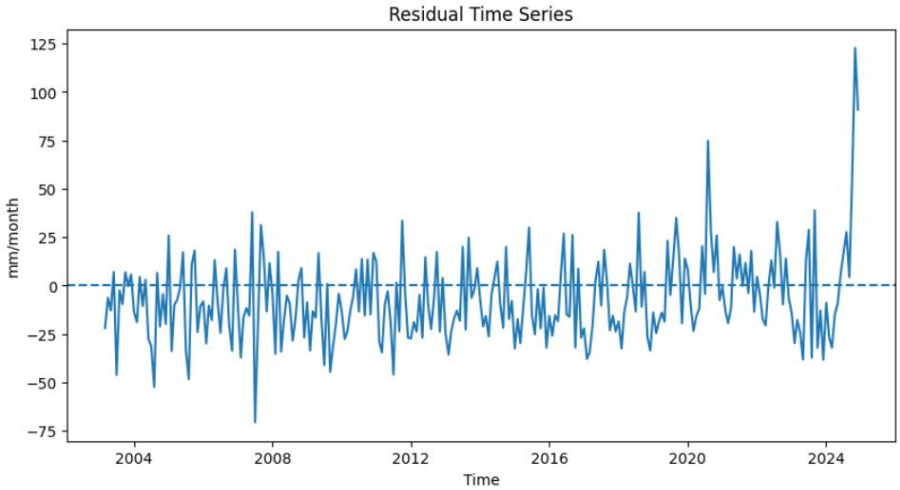
Residual = Precipitation – (Soil Moisture + Evapotranspiration + Runoff + Groundwater)

Spatial Distribution:

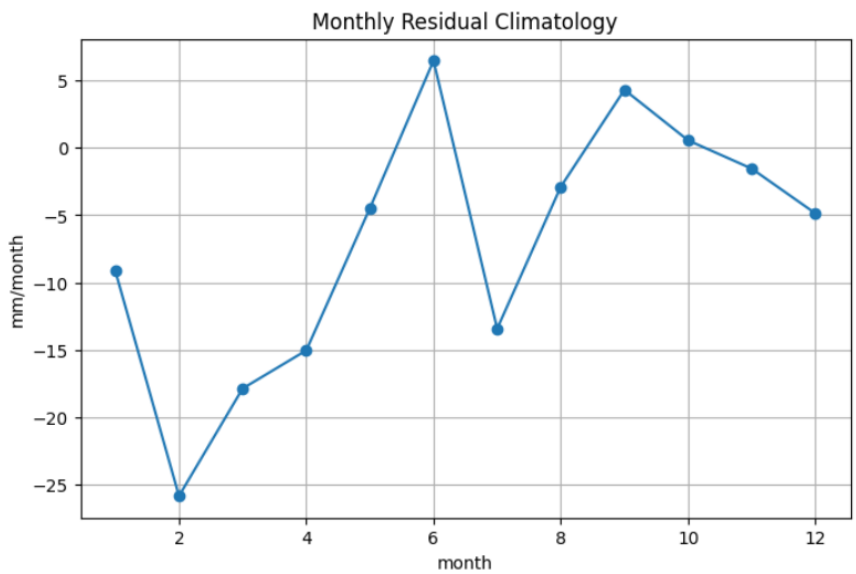


Mean residual spatial patterns show that overall, most of Peninsular India has a spatial distribution with negative values generally occurring throughout much of the western and interior regions, while most of the eastern coast has mostly positive values. The negative residuals found over the western Deccan Plateau and in much of the western region may indicate that although there are various changes being made to the hydrological cycle, such as changes to ET (Evapotranspiration), runoff, and storage, there may be an overall excess of output relative to input from these two processes combined. This is consistent with these areas being semi-arid, where hotter temperatures lead to greater ET and relatively little precipitation. The presence of negative residuals may also indicate errors in estimating ET or runoff in the available datasets. On the other hand, the eastern coast exhibits positive residuals, indicating that precipitation exceeds the combined outputs of ET and runoff. A distinct aspect of this analysis is the contrast observed between the windward side and the leeward side of the Western Ghats. The windward side receives large amounts of precipitation, whereas the leeward side has negative residuals. This indicates that orography plays a large role in controlling the regional hydrology.

Temporal Analysis :



The residual time series for the peninsular region of India show strong variability over the course of years and have generally fluctuated closely around zero on a month-to-month basis without showing an apparent trend of longer duration. These data suggest that while the overall water budget may be approximately balanced over longer periods, it still experiences substantial imbalances on shorter (e.g., short-term (monthly)) time scales. There have been many instances of the residuals being negative, showing that evaporation has frequently exceeded precipitation during the extremely dry periods. In addition, there are more extreme instances of the residual being high (positive) that can be attributed to large rain events associated with monsoons or tropical cyclones affecting the peninsula. Recent data show an increase in the amplitude of these fluctuations and an increasing frequency of extreme values of positive residuals. This may be due to an increase in the overall variability of the water budget as a result of very extreme weather during the time.



Based on observations and with reference to the above figure, the pattern of monthly residuals throughout Peninsular India appears to match the monsoon pattern. During the dry months (January through April), residuals were negative due to intense evapotranspiration with little

or no influence from rainfall. When the southwest monsoon starts in June, residuals become positive because precipitation exceeds water loss from runoff and evapotranspiration. This period is considered the "main" wet season across most of Peninsular India.

However, evidence also shows that the residuals decrease or become negative after July, even though July is typically one of the wettest months of the monsoon season. This suggests that while rainfall may provide significant water during periods of consistent precipitation, the combined runoff and evapotranspiration may offset some of those gains. By September, residuals revert to positive, then turn negative in the months following.